

February 11, 2026

Dr. Michael Hegedus
School of Engineering Science
Simon Fraser University
Burnaby, BC, V5A 1S6

Re: ENSC 405W / 440 Requirements Specification for Autonomous Catering Event & Service Robot

Dear Dr. Hegedus,

The attached document outlines a Requirements Specification for the design and implementation of an autonomous catering event and service robot, which my group and I have put together. This system aims to alleviate the middleman when moving chairs from storage to event spaces in large event facilities. A system that can autonomously travel from point A to point B indoors and transport a load on a platform could reduce repeated trips, improve turnaround time for venue setup, and reduce strain during high-volume events.

Our company aims to accomplish this goal by introducing robotics and automation to a normally manual task. We are to design a robot cart that can hold the weight of many chairs and transport them from storage to a requested location. This robot will be able to move to its requested location by itself and avoid any obstacles that may happen to get in its way. With many of these carts working in tandem, we can remove the need to have workers moving trolleys back and forth. With this we can cut down cost on labour and focus on setting up the events.

Company 10 consists of three computer engineers and one systems engineer who are all keen to explore this problem and its solutions: Mukun Chen (Jason), Rodrigo Arce Diaz, Zakrye Osmond, and Demir Cicek. As a whole, we have a greater background in hardware and software engineering, and a lot of experience in meeting the requirements of this project for us.

Thank you for taking the time to review our Requirements Specification. If you have any questions about it, please contact our Chief Communications Officer, Rodrigo Arce Diaz, by phone (778-879-1070) or by email (ria12@sfu.ca).

Sincerely,

Rodrigo Arce Diaz
Chief Communications Officer
Company 10

Enclosed: Project Proposal for Autonomous Catering Event & Service Robot

Requirements Specification (ENSC 405W)

Project: Autonomous Catering Event & Service Robot

Company: Company 10

Authors:

Demir Cicek (CEO)

Mukun Chen (CFO)

Rodrigo Arce Diaz (CCO)

Zakrye Osmond (CTO)

Date: 2026-02-11

Abstract

This document presents the requirement specifications for an autonomous indoor transport robot developed by company 10 to optimize the handling of repetitive manual transport of equipment in the catering and event industries. In large facilities, the manual transport of heavy equipment, such as chairs and tables, is a labour-intensive and repetitive task that often leads to unnecessary lost time and physical strain on staff. To mitigate these challenges, we propose an automotive robot solution performing point-to-point navigation within a controlled, single-floor indoor environment.

The technical framework is categorized into four sub-domains: Mobility and Navigation, System Control and Logistics, Payload Handling, and Remote Communication. The project follows a three-stage engineering roadmap, progressing from an initial Proof-of-Concept (PoC) to a robust Engineering Prototype (EP), and finally a Product Version Prototype (EV), indicating a market-ready design. Requirements detail key features, including a ROS2-based navigation, obstacle response process, and remote teleoperation and monitoring. By automating repetitive transport, this system aims to reduce labor costs, optimize setup times, and significantly enhance workplace safety in catering and event environments.

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Glossary

- $\geq, \leq, <, >$: Comparison operators used in requirements/metrics ($\geq$: greater than or equal to; $\leq$: less than or equal to; $<$: less than; $>$: greater than)
- **PoC:** Proof-of-Concept (Stage A)
- **EP:** Engineering Prototype (Stage B)
- **PV:** Production Version (Stage C)
- **A $\rightarrow$ B:** Point-to-point mission from origin to destination
- **ROS 2:** Robot Operating System 2
- **Nav2:** ROS 2 Navigation framework
- **SLAM:** Simultaneous Localization and Mapping
- **QoS:** Quality of Service (ROS 2 communication settings)
- **Teleop:** Teleoperation (manual remote driving)
- **LiDAR:** Light Detection and Ranging sensor
- **Encoder:** Sensor measuring wheel rotation for motion feedback
- **Hz:** Hertz

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1. Introduction & Background

This requirements specification defines an autonomous indoor transport robot that can move a payload from Point to Point in a controlled indoor environment, with safe obstacle response and operator override. Detailed requirements are organized into 4 parts: Mobility & Navigation, System Control & Logistics, Payload Handling & Secure Load, and Remote Communication & Manual Override, and are staged from PoC to Engineering Prototype to a production-oriented version.

1.1 Problem Statement

Catering and event services require frequent movement of equipment (e.g., folding chairs, bins, or supplies) between storage areas and event spaces. This work is time-sensitive, repetitive, and often done in indoor facilities with long corridors and frequent back-and-forth trips. In practice, staff commonly rely on manual carts or repeated hand-carrying. As a result, operations can become inefficient during peak periods, and workers face physical strain from repetitive manual material handling. A system that can autonomously travel from point A to point B indoors and transport a load on a platform could reduce repeated trips, improve turnaround time for venue setup, and reduce strain during high-volume events.

1.2 Proposed Solution

The proposed solution is an indoor autonomous transport robot for catering/event services, designed primarily to perform reliable point-to-point navigation between predefined locations using a ROS-based mapping and navigation pipeline to localize on a saved map and repeatedly execute route-based deliveries with minimal manual guidance[6][8]. What is newly explored in this project is an efficient and venue-oriented “route dispatch” workflow with named service waypoints and repeatable A-to-B runs tailored to event setup logistics. To keep the project feasible, we will constrain the scope to a single-floor indoor environment and an intelligent workflow, prioritizing repeatable route success over advanced features.

1.3 Stakeholders and User Roles

Stakeholder / User Role	Description	Primary goals	Key needs (inputs to requirements)
Daily User	Staff who request item delivery (e.g. event staff).	Get items delivered quickly and reliably.	Simple dispatch/booking, clear job status, ability to cancel, predictable completion time and live robot status.
Operator / Supervisor	Person who monitors missions and intervenes when needed.	Keep operations efficient and tasks successful.	Alerts when help is needed; clear recovery procedure.
Maintenance	Person responsible for setup and updates	Maintain uptime and troubleshoot failures.	Logs and fault codes; testable subsystems; configuration-based setup; repeatable calibration steps.
Facility Management	Stakeholder responsible for workflow, safety, and asset loss.	Reduce labor costs and minimize incidents.	Safety behavior; mission history; clear limitations and operating scope; reliability metrics.
Bystanders (Indirect Users)	People sharing the indoor space with the robot.	Stay safe and comfortable around the robot.	Conservative speed; obstacle stop; visible “moving/stopped” indication.

Table 1: Stakeholders and User Roles

14 Scope and Out-of-Scope

In Scope

- Indoor, single-floor operation in hallways/rooms on smooth floors
- Autonomous point-to-point travel
- Transport of defined payload types within stated payload limits
- Obstacle response and safe-stop on faults
- Operation for system control, basic status, and scheduling

Out-of-Scope

- Outdoor terrain, stairs, or unknown surfaces
- Fully unsupervised operation in crowds
- Object manipulation or handling liquid

2. High-Level System Description and Requirement Outlines

2.1 Requirement Overview

This indoor mobile consists of a software system, onboard ROS2 compute, edge-device (Raspberry Pi5), sensors, navigation and state management, motion control, and logging functions[6].

Therefore, requirements are organized into: functional navigation, safe actions, performance reliability, maintainability, interfaces, and verification tests aligned to the different stages.

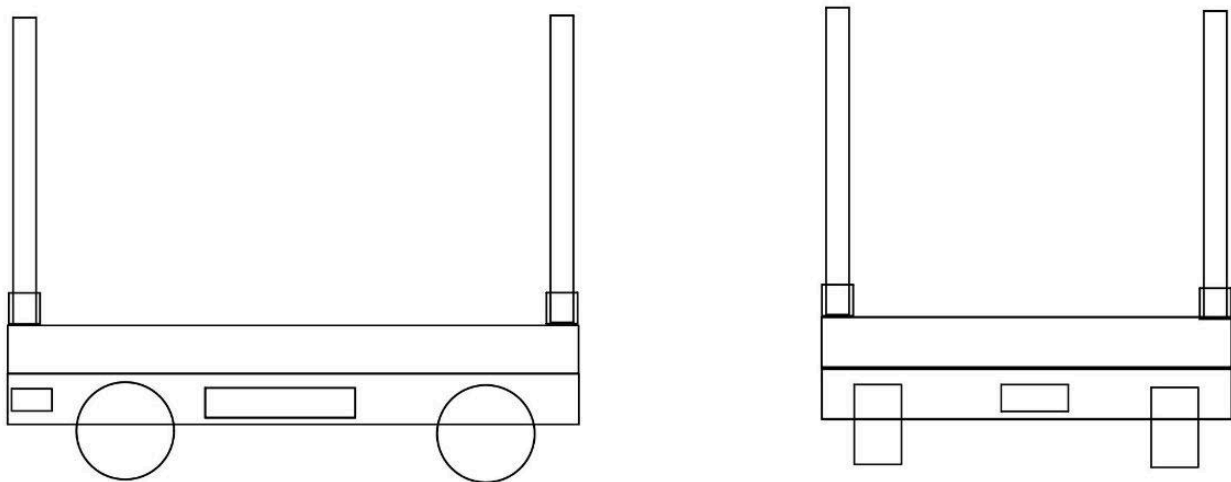


Figure 1:
DRAFT - Autonomous Catering Events and Services Robot

2.2 Proof-of-Concept Functional Block Diagram

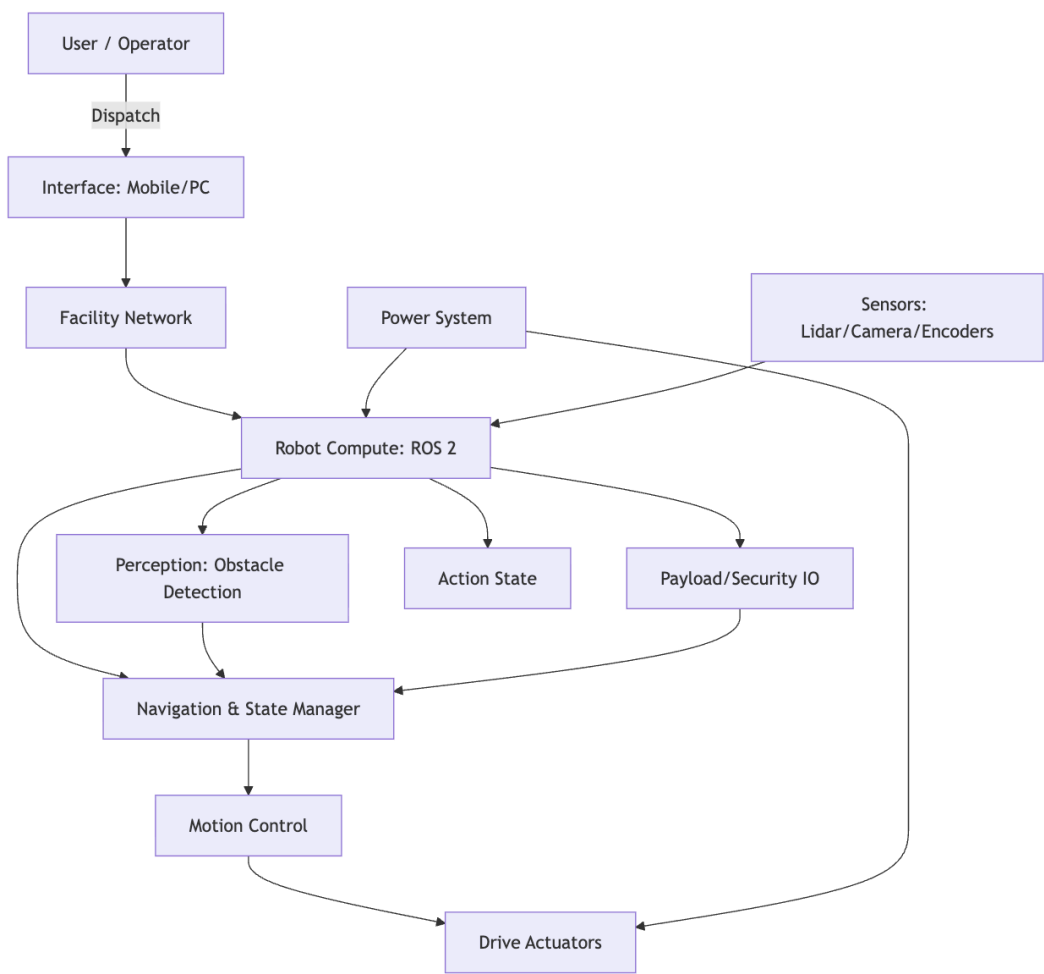


Figure 2:
Function block diagram of proof-of-concept system

2.3 Three-Stage Outlines (PoC vs EP vs PV)

Outline A — Proof-of-Concept (PoC) stage

The team will build a basic indoor transport robot that demonstrates the feasibility of the core workflow. The PoC will show simple autonomous point-to-point travel between two marked locations, basic response, payload surrogate transport without tipping, and manual override for safe operator intervention[9].

Outline B — Engineering Prototype (EP) stage

The system will be improved into an integrated prototype with more measurable performance targets. The robot will operate in a mapped indoor environment with improved navigation and obstacle handling, provide secure load transport for a chosen payload type, and include scheduling policies with correct status reporting[6][7]. Recovery behaviors such as pause, reroute, request help, and manual override will be incorporated to handle realistic hallway conditions.

Outline C — Production Version (PV) stage

The design will be refined toward a product-ready version, emphasizing robust security, higher uptime and maintainability, and a cleaner, deployment-oriented implementation. This stage also identifies future expansion items for broader facility operation as a longer-term scope.

2.4 High-Level Hierarchical Requirement Outline (what/shall/should)

Tag legend

- A: Proof-of-Concept requirements
- B: Engineering Prototype requirements
- C: Production Version requirements
- E#: Electrical requirement
- M#: Mechanical requirement
- S#: Software requirement

Key Requirements

Mobility & Navigation:

- Needed so the robot can move to destinations indoors on the map while avoiding collisions and operating safely.

System Control & Logistics

- Needed to turn requests into a managed mission (states, validation, status reporting) so the operation is predictable and traceable.

Payload Handling & Secure Load

- Needed to ensure the robot can carry and retain the load safely without slipping, tipping, or dropping during motion.

Remote Communication & Manual Override

- Needed for supervised operation, enabling teleop, live status, and an immediate stop and override for safety.

3. Detailed Requirements

System Overview:

The overall system will have a robot dedicated to moving loads from one area to another and an interface system dedicated to controlling the robot remotely. The robot should handle a significant load that can be used for moving multiple chairs and have a manual override in case problems arise.

Main system requirements overview & Assignment

3.1 Mobility & Navigation (owner: Rodrigo)

Function 1: The robot must be able to navigate around obstacles

3.2 System Control & Logistics (owner: Mukun)

Function 2: Mission management in decision logic and job scheduling

Function 3: System interfaces & configuration

3.3 Payload Handling & Secure Load (owner: Demir)

Function 4 - System Integrity and Load Stability

3.4 Remote Communication & Override (owner: Zak)

Function 5 - Teleoperation & Supervisory Control

3.1 Mobility & Navigation

-Author:Rodrigo Arce Diaz

The robot and its subsystem has to accomplish many autonomous movement capabilities within indoor environments. It is responsible for the building and use of maps, determining its own position, planning best routes to its destination, detecting obstacles, and controlling wheel motion. This system ensures safe, smooth travel that can work even in tight spaces, while following commands provided by administrative control software.

Function 1: The robot must be able to navigate around obstacles			
Tag	sub -function	Description	Justification
A	S3.1.1	The robot shall be able to generate or load a single map for use during navigation.	The main mobility & navigation goals that the robot needs to accomplish for the proof of concept are the generation of a map, detection of obstacles, and movement in every direction. These requirements that are listed cover all those necessary goals. Having the map allows the robot to differentiate between map and obstacle and allows it to stop before hitting it. Having control of the wheel speed allows us to move in every direction with manual control.
	S3.1.2	The robot shall estimate its position on a map well enough to reach a goal.	
	S3.1.3	The robot shall be able to detect an obstacle in front of it and stop before it makes any contact	
	M3.1.4	The robot shall control the wheel speeds using encoder feedback	
	M3.1.5	The robot needs to be able to move with manual control, forwards, backwards, turn left, and turn right	
B	S3.1.6	The system shall build a consistent 2D map of the operating area	The main mobility & navigation goals that the robot needs to accomplish for the engineering prototype stage are goals regarding autonomous use of the robot. The robot needs to be given a goal and be able to autonomously navigate to it using its map data while continuously updating routing and avoiding collisions.
	S3.1.7	The system shall plan the route to destinations from mission software.	
	S3.1.8	The system shall compute collision free paths.	
	S3.1.9	The system shall localize and follow paths with less than 10 cm deviation.	
	S3.1.10	The system shall detect both static and moving obstacles	
	S3.1.11	The system shall re-route or stop when required and resume when a path clears or a new path is computed.	
	M3.1.12	The robot shall reduce speed when obstacles are near or having to navigate through tight spaces including doors	
	M3.1.13	The robot shall execute smooth arc and in-place turns depending on need at time	
C	S3.1.14	The system shall support long-term map reuse and new map loading	Production is justified because it enables deployment across different environments. Using a preloaded map removes the need for time-consuming environment discovery. Omni-directional mobility provides more flexible maneuvering than standard wheels, improving positioning and storage efficiency.
	M3.1.15	The robot will have omni-directional movement	

Table 2: Mobility & Navigation Requirements

3.2 System Control & Logistics

-Author: Mukun Chen

System Control & Logistics is the software layer that turns dispatch requests into organized robot missions. It manages scheduling, mission states, and logging for traceability. **Metrics** are added here because these behaviors must be testable, so that metrics make “works correctly” measurable. The functionality has split into two functions: Function 2) **Mission management in decision logic and job scheduling**; Function 3) **System interfaces & configuration**. **Figure 3** (Intended only to assist users with visualization) shows a concise process for system control of the robot, providing a solid base for further improvement and extension.

Function 2: Mission management in decision logic and job scheduling

Stage	sub -function	Description	Justification
A	S3.2.1	The system shall implement an action state control with Idle, En-route, Paused, and Completed states.	The action state control is the minimum structure needed to track one trip end-to-end and to show a consistent workflow during the PoC demo [1][2].
	S3.2.2	The system shall identify and validate a mission request specifying a destination in the loaded map.	Mission request validation ensures the system can accept commands from an external user platform reliably; it keeps Stage A realistic and testable.
	S3.2.3	The system shall enforce a manual safe pause mode (operator-triggered) that stops motion and pauses mission progress.	Manual safe pause supports safe human intervention during PoC without requiring full autonomous recovery [5].
B	S3.2.4	The system shall support a clear running schedule(e.g., single/round trip) from action state control using policies to prevent conflicting missions.	The schedule prevents conflicting or incorrect missions and defines behavior under contention (reject/queue/reschedule), which is necessary before usage.
	S3.2.5	The system shall detect “blocked/trapped” signal while En-route	Blocked/trapped detection reduces failure rate in real hallways where temporary blockage is common; the retry limit prevents unsafe endless behavior and escalates to operator help[4].
	S3.2.6	The system shall log a mission record: ID, destination, timestamps, and outcome.	Mission logging enables debugging, test evidence for verification, and supports later evaluation.
C	S3.2.7	The system should support scaling to multiple robots (fleet) without changing the mission request interface (same schema).	Fleet scalability is intentionally Stage C because it is a long-term capability beyond the PoC/EP demo scope.

Table 3: System Control & Logistics (Mission Management) Requirements

Metric & Type

S3.2.1 [Functional]: States observed across ≥ 1 mission

S3.2.2 [Functional]: $\geq 90\%$ valid requests accepted; $\geq 90\%$ invalid destinations rejected

S3.2.3 [Functional]: Pause takes effect within ≤ 2 s

S3.2.4 [Functional]: Conflicts handled correctly $\geq 90\%$;

S3.2.5 [Functional]: Condition detected by ≤ 5 s; recovery begins ≤ 2 s after detection; help request issued after ≤ 3 retries

S3.2.6 [Functional]: $\geq 90\%$ missions logged; record contains all required fields

S3.2.7 [Non-Functional]: Interface unchanged; multiple robot IDs supported in mission model

3.2 System Control & Logistics

-Author: Mukun Chen

Function 3: System interfaces & configuration			
Stage	sub -function	Description	Justification
A	S3.2.8	The system shall provide a status interface exposing the task state and destination.	Status interface and subsystem inputs are required to observe and control the robot's mission progress and integrate essential flags and information for further work (localization valid, obstacle stop, etc.).
	S3.2.9	The system shall provide the interfaces to receive user and external subsystem status inputs required for mission control (e.g., localization/stop).	
B	S3.2.10	The system shall support config-driven destinations: adding/removing destinations shall be possible through configuration rather than code changes.	Config-driven destinations and sim/hardware consistency make it easier to expand from PoC to EP without rewriting code and enable early validation in simulation using the same external interface[3].
	S3.2.11	The system used for mission requests and status shall be consistent between the simulation and the physical robot to enable validation of the robotic configuration.	
C	S3.2.12	The system should log command-source metadata via the interface (who/what issued dispatch/cancel/abort) for mission-control commands that occur.	Command-source logging supports auditability and analysis in later stages.
	S3.2.13	The system should provide a convenient interactive interface, such as natural language manipulation and object recognition	Scalable multi-interaction functions will endow robots with more capabilities

Table 4: System Control & Logistics (System Interfaces & Configuration) Requirements

Metric & Type

S3.2.8 [Functional]: Status refresh ≥ 1 Hz

S3.2.9 [Functional]: Receives ≥ 3 status signals without error during ≥ 1 mission

S3.2.10 [Functional]: Config change takes effect after restart/reload; 0 code changes

S3.2.11 [Non-Functional]: Same request/status schema in sim and hardware; same test script works in both

S3.2.12 [Non-Functional]: 100% of mission-control command log entries include a command source field

S3.2.13 [Functional]: success rate $>80\%$ for natural language control and object recognition

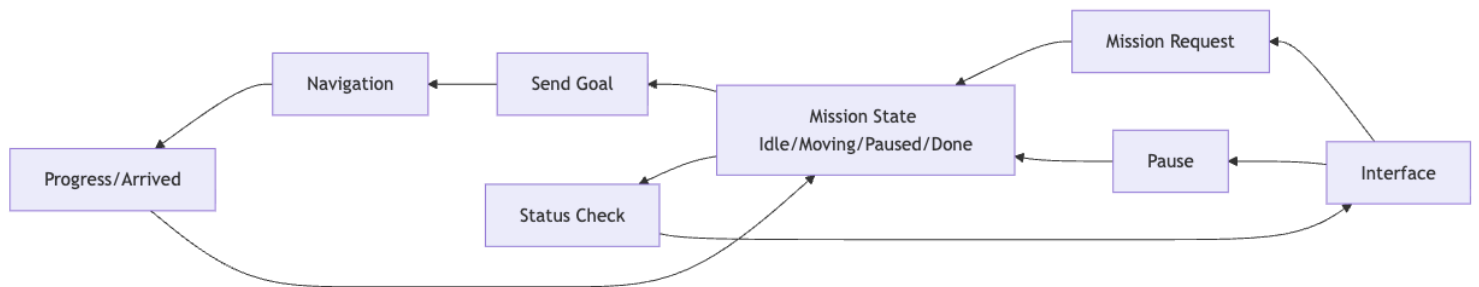


Figure 3:

This diagram is a conceptual illustration of a control flow, not a complete design.

3.3 Payload Handling and Secure Load

-Author:Demir Cicek

The transport robot must be capable of safely carrying and transporting payloads of up to **50kg**, with an assumed maximum system load of **~100kg** for safety margin. The robot will prevent the payload from shifting during operation, including acceleration, deceleration and turns. The system should also consider supporting a payload locking mechanism to improve safety and reliability.

Function 4 - System Integrity and Load Stability			
Stage	sub -function	Description	Justification
A	M-3.3.1	The robot shall be capable of supporting and transporting a payload of 50kg without structural failure.	At this stage, structural integrity and basic load stability are essential to allow safe operation and validate that the chassis and load platform are capable of performing the expected task. The focus will be on preventing major shifting during controlled motion using simple retention methods.
	M-3.3.2	The load platform shall prevent major payload shifting during basic forward acceleration on flat indoor flooring.	
	M-3.3.3	The robot shall maintain payload and chassis stability during low-speed turns and deceleration under testing conditions.	
	M-3.3.4	The system should include a basic payload retention method to reduce the likelihood of shifting. (Ex:Boundary around edge, surface with high friction, straps or other restraints.)	
B	M-3.3.5	The robot shall transport a 50kg payload through typical indoor routes such as straight paths, turns, stops, without the payload shifting outside the designated cargo region.	In a typical facility the robot will encounter turns, repeated stops and floor irregularities such as seams, bums, dips etc. Without improved retention, accumulation of small load shifts can result in hazards such as loss of load, collisions and general instability. With improved retention along with additional software and electrical requirements confirming load presence and retention engagement risk is reduced. Enforcing safe behaviour before motion can occur.
	M-3.3.6	The load platform shall be designed to prevent payload sliding during expected acceleration and deceleration events, simulating normal operation conditions. This includes sudden emergency stop scenarios.	
	M-3.3.7	The robot will maintain retention while traversing floor anomalies such as seams, thresholds, bumps, cables or small dents which can typically be found in indoor facilities.	
	M-3.3.8	The system should include an optional mechanical lock to secure loads during transport when required.	
	S-3.3.9	The robot will prevent motion from occurring if the payload retention system has not been properly engaged.	
	E-3.3.10	If an active retention system is used, an electrical interface to enable and disable the retention mechanism. The robot shall provide a clear indication (visually and/or audibly) if the payload has been properly engaged.	

C	M-3.3.11	Any implemented paylock locking/retention mechanism shall be operable by a single user in less than 10 seconds and will not require any tools for normal operation.	The focus of the production version is to ensure the system is safe, reliable and practical for real world use. In operational environments, payloads may be loaded and unloaded frequently, the user must be able to quickly do so without any tools. Components are expected to wear down over time or receive unexpected damage. Exposed key components must be easily replaceable without major disassembly to minimize downtime and reduce expertise for repair. A modular retention system allows accommodation of new cargo, increasing market scalability.
	M-3.3.12	Components that can be easily damaged or are prone to wear should be easily replaceable without requiring disassembly of the chassis or any other major components.	
	M-3.3.13	The system will include a modular load retention system method capable of supporting additional cargo types.	
	S-3.3.14	The robot will provide a clear indication of payload secure status prior to and during motion using some sort of indicators on the robot and potentially on any device used to operate it.	
	S-3.3.15	The system should be able to confirm the presence of a payload either through manual input or automatically via sensor. Blocking operation if a load is present without all safety engaged.	
	S-3.3.16	The robot shall cease all operations if the payload weight drops past an acceptable threshold indicating something may have fallen, issuing an alert to the user.	

Table 5: Payload Handling & Secure Load Requirements

The payload handling requirements in the project stem from functionality driven needs as well as established safety and load handling standards. The ISO 10218-1:2025 provides design principles for secure load handling and risk reduction during robot maneuvering; preventing loss of load from unintended movement which could otherwise endanger nearby individuals. While the ISO 10218-1 is primarily focused on industrial robotics, it presents clear core principles on load retention, status indication and mechanical integrity which are directly applicable to a mobile transport assistance robot.

3.4 Remote Communication & Override

-Author: Zakrye Osmond

The Transport robot will be operating at a distance and must be able to be accessed by users from any location. The user will be able to intervene and take direct remote control of the robot through a Wi-Fi connection. Allowing the user to control, command, or remotely stop the robot's action. The user will be able to access and monitor the robot's status, telemetry, and battery through a visual interface. The robot will be able to perform a safe shutdown remotely through the interface.

Function 5 - Teleoperation & Supervisory Control			
Stage	sub -function	Description	Justification
A	S-3.5.1	The system shall communicate with the user over a point-to-point Wi-Fi connection.	A wireless connection is required to allow the user to have control and communicate with the robot.
	S-3.5.2	The system shall allow the user to remotely give directional instructions for basic movements (Forward, Backward, Left, Right)	Communication of manual movement instructions is required to allow the user to be able to initiate manual intervention and perform all desired movements.
	S-3.5.3	The user shall be able to execute a "Stop" action that overrides all movement instructions immediately.	A system movement override function is crucial for critical safety requirements and allows the user the ability to intervene during any unforeseen behaviour or situations. [10]
	S-3.5.4	The robot shall communicate a real-time feedback output of its current battery voltage, and motor state to the user interface at a rate of 1Hz.	Feedback information is required to allow the user to get insight into internal operations and have trust in the system. Essential for ensuring proper operation, testing, and debugging. [11]
	S-3.5.5	The system shall display the connection status and indicate on the user interface if the connection is lost.	Connection status check is required to allow the robot to communicate when instructions are received or connection has been lost.
	S-3.5.6	The user shall be able to perform a remote software reboot or safe shutdown command through the Wi-Fi communication.	Safe shutdown and reboot ensures the sustainability of the robot and its parts, allowing the robot to have a normal state on startup.
	S-3.5.7	The robot shall accept destination point locations through wireless communication.	The users require the ability to provide location instructions to wirelessly communicate the destination point to the robot.
B	S-3.5.8	The system will communicate over a roaming Wi-Fi connection.	A roaming connection allows the robot to communicate while navigating through different access points to sustain connection through deadzones.
	S-3.5.9	The robot shall communicate real-time feedback including status, current coordinates, and time to destination to the user interface at a rate of 5Hz to 10Hz.	Real-time feedback allows the user to gain insight on the location and operation of the robot at all times.
	S-3.5.10	The system shall indicate through a visual alert on the user interface when	When switching to manual override, an alert is required to notify the user that the manual control is

		a manual override is active.	active and operating correctly.
	S-3.5.11	The system shall display the current battery voltage and an estimated remaining percentage, will trigger a “Low Power” alert through the interface once the battery reaches below 15%.	Low power or battery alerts notifies the user of the robot's remaining working capacity and reduces the risk of safety concerns and hazards when completing a task.
C	S-3.5.12	The system shall communicate over Wi-Fi and will have a 4G/5G Cellular failsafe.	4G/5G failsafe allows the robot to have constant connection while navigating through Wi-Fi deadzones. [12]
	S-3.5.13	The system’s remote access and data shall be encrypted using a VPN.	Data being transferred wirelessly can be accessed by unauthorized users and requires a VPN to ensure security and safety.
	S-3.5.14	The system shall include remote Firmware Updates through the user interface.	When firmware updates are required, the user must be able to implement them wirelessly.
	S-3.5.15	The system shall incorporate a cloud-based infrastructure.	Cloud-based infrastructure will grant authorized users the opportunity to remotely access and monitor the robot. Eliminating the local network barrier while increasing convenience and efficacy.

Table 6: Remote Communication & Override Requirements

4. PoC Completion Plan

4.1 Objective

The objective of the PoC stage will be to demonstrate that the transport robot is capable of completing a basic payload delivery mission from point A to point B in a controlled indoor environment while meeting all the core functional requirements. This stage will prioritize basic navigation, mobility, and payload stability.

PoC deliverables require coordinated software, electrical, and mechanical work, and subsystem responsibilities are summarized in Section 4.3.

4.2 PoC Deliverables – What we will produce

ID	Description
4.2.1	Mobile platform capable of controlled forward (optional reverse) motion and turning capabilities, whilst avoiding collision with the environment
4.2.2	Basic point-to-point movement in a controlled indoor environment. The operator-selected destination will determine waypoint pathing.
4.2.3	Obstacle detection, avoidance, and safe stop capability. The system shall prevent forward motion when an object is detected within a defined stopping distance. This will also test the response to dynamic obstacles and the response time of the system.
4.2.4	Stable operation while supporting the anticipated total system mass of 100 kg. (Including the 50 kg payload). The load shall remain on the platform during all forms of movement.
4.2.5	Basic remote interface allowing the user to manually drive the robot using directional controls. Remote communication shows the current status and is able to execute the request/stop commands.
4.2.6	Functional test results with a video demonstration. This will include a demo of both manual and automatic pathfinding with at least one full A to B run.

Table 7: PoC Deliverables

4.3 PoC Work Breakdown – How we will get there

Mobility & Navigation Select the drive configuration and motor control interface, implement basic localization and waypoint-based A → B navigation, and integrate obstacle detection with safe stop behavior. Model the robot for software simulations and proper navigation capabilities.	System Control & Logistics Provides the minimum “mission backbone” needed to run a safe, observable demo: it accepts and validates a destination request, drives a simple mission state model (Idle, En-route, Paused, Completed), supports an operator-triggered safe pause that stops motion immediately, and publishes a basic status interface while ingesting essential subsystem signals for mission control decisions.
Payload Handling & Secure Load Define payload platform requirements and test setup, implement load stabilization (slippage, boundaries, restraint). Validate payload retention during straight driving, turning and braking. Document safe payload handling procedures for PoC demo	Communication & Manual Override Implement remote telemetry and system status reporting, manual teleoperation controls, remote stop override, and battery monitoring with a low-battery warning.

5. Engineering Prototype Completion Plan

5.1 Objective

The objective of the Engineering Prototype (EP) stage is to deliver an integrated indoor transport robot that can reliably run repeatable A to B missions on a saved map, handle typical obstacles and blockage recovery, carry a chosen payload safely, and provide robust remote monitoring/control with measurable performance targets.

EP deliverables are system-level outcomes requiring coordinated software, electrical, and mechanical work, and subsystem responsibilities are summarized in Section 5.3.

5.2 EP Deliverables – What we will produce

ID	Deliverable (EP)
5.2.1	Mapped navigation: Mapped navigation builds a consistent map, accepts destinations from the mission software, computes collision-free paths to the goal, and keeps behavior consistent between simulation and the real hardware.
5.2.2	Obstacle behavior: The system must detect static obstacles, reduce speed when operating near obstacles, and stop or maneuver to avoid contact.
5.2.3	Recovery behaviors: When blocked, the robot must either reroute or perform a safe stop-and-resume, using defined trapped-response policies when it cannot proceed.
5.2.4	Mission control: The mission software must enforce a scheduling policy to prevent conflicts, detect blocked/trapped conditions, record mission logs, use configuration-driven destinations, and maintain consistent system communication across components.
5.2.5	Payload security (50 kg): The payload must remain within the defined cargo region during straight motion, turns, stops, and typical floor anomalies, with an optional locking mechanism and payload-presence confirmation if those features are implemented.
5.2.6	Remote and communication: The robot must support roaming Wi-Fi across access points and use a connection watchdog that stops the robot if Wi-Fi is lost for more than 2 seconds.
5.2.7	Manual override: The system must support teleoperation with ≤ 200 ms latency, provide a remote stop function, and allow remote reboot, shutdown, and reset.
5.2.8	Telemetry: The system must report status, coordinates, and battery voltage/percentage, and issue a “Low Power” alert.

Table 8: EP Deliverables

5.3 EP Work Target Breakdown – How we will get there

Mobility & Navigation:

- map and localization tuning
- collision-free planning
- moving obstacle handling
- speed reduction near obstacles
- smooth turns; path tracking to <10 cm.

System Control & Logistics:

- optimized scheduling policy
- blocked/trapped detection integration
- retries and help requests
- mission logs
- config-driven destinations

Payload Handling:

- solid platform and retention
- validate 50 kg through typical routes, validate on floor anomalies
- optional lock and payload presence reporting.

Communication & Manual Override:

- roaming Wi-Fi
- connection-loss stop
- telemetry improvement
- battery % alert
- remote reboot/shutdown/reset.

Appendix — Proof-of-Concept Poster Demonstration

This appendix identifies the minimum PoC functionalities that will be demonstrated during the ENSC 405W poster session for the autonomous indoor transport robot.

PoC Demonstration

A.1 Mobility & Navigation (PoC) • Demonstrate controlled mobile base motion for forward movement and turning in a constrained indoor area. • Demonstrate basic point-to-point travel to an operator-selected destination in a controlled environment. • Demonstrate obstacle response and safe-stop behavior such that forward motion is prevented when an obstacle is within the defined stopping distance.
A.2 System Control & Logistics (PoC) • Demonstrate core mission logic that transitions through a clear mission flow, such as idle to moving to completed. • Demonstrate a minimal system interface, where mission requests and status follow a defined schema. • Demonstrate a fundamental status display that reports, at a minimum, the robot’s current state and destination during operation.
A.3 Payload Handling & Secure Load (PoC) • Demonstrate transport of a payload surrogate such that the load remains on the platform throughout motion with no tipping and off-platform loss under the expected PoC driving conditions. • Demonstrate stable operation while supporting the expected PoC system mass, including a representative payload.
A.4 Remote Communication & Manual Override (PoC) • Demonstrate basic remote communication for manual driving and directional control over Wi-Fi. • Demonstrate the pause command that overrides motion immediately. • Demonstrate telemetry and status reporting to the user interface, including battery voltage display, if available.

Table 9: PoC Poster Demonstration Functionalities

Key Problems & Mitigations

Key problem (PoC)	Why it matters for the demo	Mitigation (PoC only)
Power instability and unexpected reset	Demo fails mid-run; unsafe behavior risk	Fully charge battery; secure connectors; run a test with payload; keep speeds conservative
Manual override is not reliable	Safety risk; can’t recover from bad behavior	Treat the stop as the highest priority; test the stop repeatedly; keep the operator within reach to cut power if needed
Obstacle detection is not consistent	Collision risk; fails the “safe-stop” moment	Use a conservative stop distance; slow down near obstacle zones; be ready to stop manually
Wi-Fi drop and control latency spikes	Loss of control or action freezes during the demo	Use a simple, close-range network setup; keep the robot within strong signal range; have a clear manual-stop fallback
Payload shifts or tips	Demo looks unsafe and unreliable	secure payload surrogate; limit acceleration and turning speed; validate payload stability
Mission state not matching robot behavior	Audience confusion; hard to show “mission flow”	Keep mission states clear from Idle to Moving to Completed; update state+destination at a steady rate
Loose mechanical parts	Sudden failure; safety issue	inspection checklist for fasteners, wheels, cables; tidy cable routing; short test drive before presenting

Table 10: Key Problems and Mitigations (PoC)

Conclusion

This document defines staged requirements for Company 10's indoor autonomous facility logistics robot. The PoC focuses on demonstrating feasibility in navigation, reliability, and intelligence, while the Engineering Prototype targets measurable performance with a chosen payload and scheduling functionality. The Production Version outline identifies future expansion areas in robust security, deployment readiness, and scalability.

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